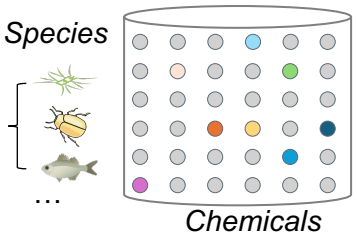
1. Broad Motivation

Sparse toxicity tests



Most species-chemical combinations remain unmeasured.

2. This Study Design

Inputs

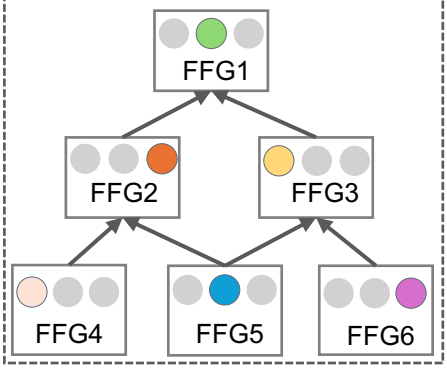
- Sensitivity
- Body mass
- Functional feeding group
- Phylogeny
- Chemical-use group

Outputs

- Phylogenetic signal
- Structured sensitivity patterns
- Uncertainty

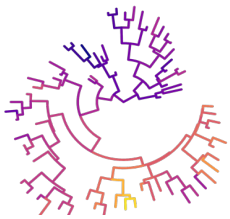
3. Future Application

Potential integration into food web risk assessment



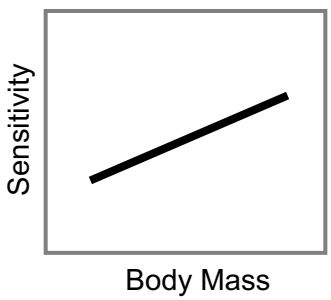
Step 1. Quantify phylogenetic signal

Quantify phylogenetic signal in species sensitivity within each taxonomic group for each chemical

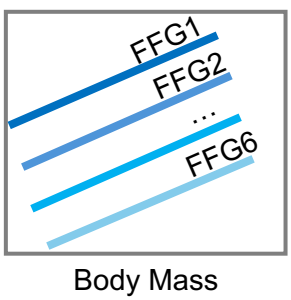


Step 2. Test Bayesian hierarchical models with explicit phylogenetic effects

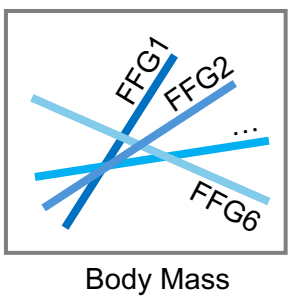
H1: Global Allometry



H2: FFG Baselines

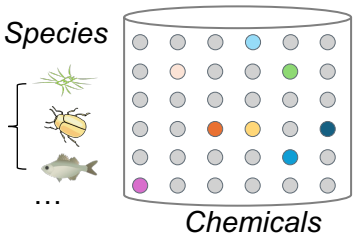


H3: FFG Allometry



1. Broad Motivation

Sparse toxicity data



Most species-chemical combinations are untested.

2. This Study Design

Inputs

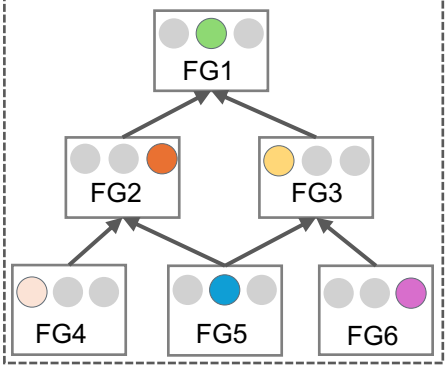
- Sensitivity
- Body mass
- Functional group
- Phylogeny
- Chemical-use group

Outputs

- Phylogenetic signal
- Structured sensitivity
- Uncertainty

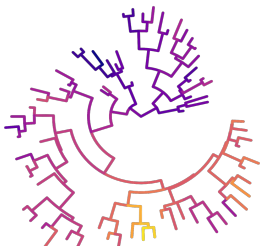
3. Future Application

Integration into food web risk assessment



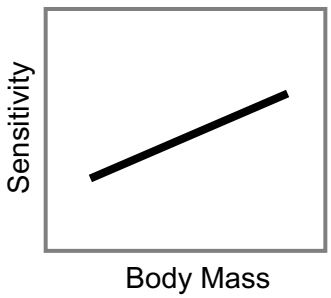
Step 1. Quantify phylogenetic signal

Estimate phylogenetic signal in sensitivity for each chemical

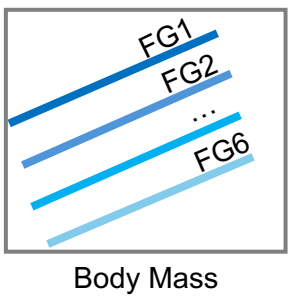


Step 2. Fit Bayesian phylogenetic hierarchical models

H1: Global allometry



H2: FG-specific baselines



H3: FG-specific allometry

